Comparative Analysis of Global E-Waste Regulations

The following table provides a comparative analysis of the e-waste regulation protocols across five distinct global frameworks: India, the United States, Australia, Russia, and Japan. This study contrasts their approach to Extended Producer Responsibility (EPR), compliance pathways, financing structures, covered scope, and enforcement mechanisms based on the five protocols identified in the Russian regulatory model.

Country	Core Regulatory Framework	Compliance Mechanisms	Financial & Fee Structure	Scope & Categorization	Reporting & Enforcement
Russia	EPR Mandate embedded in broader waste management laws (Federal Law No. 89-FZ).	Flexible pathways: Independent infrastructure, direct contracting with operators, or joining industry associations.	**Eco-fee system:** A penalizing environmental tax calculated by mass/unit if operators fail to meet recycling standards.	Categorized by HS codes (computers, communications, appliances, batteries) with specific annual percentage targets.	Strict annual electronic declarations submitted to Rosprirodnadzor by April 1st.
India	E-Waste (Management) Rules, 2022 enforcing a comprehensive Extended Producer Responsibility (EPR) framework.	Mandatory registration for PIBOs; compliance achieved by purchasing EPR credits from authorized recyclers.	Market-based EPR credit exchange portal. Environmental compensation (penalties) are levied for target shortfalls.	Broad coverage (IT, telecom, consumer goods) with escalating annual collection targets (e.g., 80%).	Mandatory digital filing of quarterly and annual returns via a centralized CPCB portal.
United States	No unified federal EPR law (RCRA has limited scope). Operates on highly fragmented state-level regulations.	Varies drastically. Over 25 states have independent laws; most utilize a producer responsibility approach.	State-dependent. Mandated manufacturer-funded recycling or Advanced Recovery Fees (e.g., California) at point of sale.	Fragmented definitions. Scope restricted to specific display devices, computers, or TVs depending	State-by-state enforcement with no centralized federal tracking or reporting system.

Country	Core Regulatory Framework	Compliance Mechanisms	Financial & Fee Structure	Scope & Categorization	Reporting & Enforcement
				on the state.	
Australia	National Television and Computer Recycling Scheme (NTCRS) backed by the Product Stewardship Act.	Liable manufacturers must join government-approved "co-regulatory arrangements" to manage recycling.	Industry-funded model where liable companies pay for downstream collection; free for households.	Targeted scope focusing primarily on televisions, computers, printers, and computer peripherals.	Regulated by the Australian Government (DCCEEW) to meet specific national material recovery targets.
Japan	Home Appliance Recycling Law (2001); Small Electronic Devices Recycling Act (2013).	Shared responsibility: Retailers take back appliances, manufacturers recycle, and municipalities collect small devices.	**Consumer pays:** Consumers pay collection, transport, and recycling fees at time of disposal for large appliances.	Bifurcated system separating large household appliances (TVs, fridges, ACs) from smaller WEEE.	Highly regulated physical flow using a strict recycling ticket (manifest) system to track end-of-life channelization.